Computer Exercise 0: Practice solutions

Algorithms, Programming, and Data Representation · Module 92313 · Exercise 0

From Fundamentals to AI Apocalypse · personal study guide

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Made by Thanate Amornjiraporn (Focus)

Practice · not course-graded

Exercise 0 is a practice workflow check. Use it to verify your submission setup; it is not part of the course grade. Try each task before opening the worked steps.

Task 1: Return the exact greeting

Connected topic: Computers, programs, and data representation (Chapter 2)

Task 01 · hello()

Try first: write the function first, then compare its returned string character for character.

Answers: see "Task 1: Return the exact greeting" under Answers at the end.

Task 2: Add two numbers

Connected topic: Computers, programs, and data representation (Chapter 2)

Task 02 · add_two_numbers(a, b)

Try first: identify the return type implied by the examples and consider negative and decimal inputs.

Answers: see "Task 2: Add two numbers" under Answers at the end.

Task 3: Square a number

Connected topic: Computers, programs, and data representation (Chapter 2)

Task 03 · square(number)

Try first: express "square" as multiplication and check a negative input.

Answers: see "Task 3: Square a number" under Answers at the end.

Task 4: Test whether a number is even

Connected topic: Computers, programs, and data representation (Chapter 2)

Task 04 · is_even(number)

Try first: use the remainder after division by two.

Answers: see "Task 4: Test whether a number is even" under Answers at the end.

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Task 5: Build a personalized greeting

Connected topic: Computers, programs, and data representation (Chapter 2)

Task 05 · greet(name)

Try first: use string formatting and preserve the supplied name exactly.

Answers: see "Task 5: Build a personalized greeting" under Answers at the end.

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Answers

Task 1: Return the exact greeting

Task 01 · hello() · worked solution

1. The contract is a zero-argument function returning one exact string. There is no input to transform.
2. Return the required literal; do not `print` it, because printing produces `None` as the function result.
3. Call the function and compare the value, including capitalization, comma, and exclamation mark.
4. **The whole function:**

```
def hello():
  return "Hello, World!"
```

Task 2: Add two numbers

Task 02 · add_two_numbers(a, b) · worked solution

1. The required operation is ordinary addition. Keep the parameters generic so Python preserves its numeric behavior for integers and floats.
2. Return `a + b`; avoid converting to `int`, which would discard a fractional part.
3. Check that `add_two_numbers(2, 3)` returns 5, then check a negative value and a float.
4. **The whole function:**

```
def add_two_numbers(a, b):
  return a + b
```

Task 3: Square a number

Task 03 · square(number) · worked solution

1. A square is the number multiplied by itself; it is not the square root and it is nonnegative for real inputs.
2. Return `number * number`.
3. Check `square(4)` = 16 and `square(-3)` = 9.
4. **The whole function:**

```
def square(number):
  return number * number
```

Task 4: Test whether a number is even

Task 04 · is_even(number) · worked solution

1. An integer is even exactly when its remainder modulo 2 is zero. Python's modulo rule also handles negative integers consistently.
2. Return the Boolean comparison `number % 2 == 0`. Do not return the strings `"True"` or `"False"`.
3. Check 4 gives `True`, 5 gives `False`, and -2 gives `True`.
4. **The whole function:**

```
def is_even(number):
  return number % 2 == 0
```

Task 5: Build a personalized greeting

Task 05 · greet(name) · worked solution

1. The output has a fixed prefix, the name, and a final exclamation mark. No punctuation is added to the name itself.
2. Use an f-string to construct and return the full text.
3. Check `greet("Ada")` returns exactly `"Hello, Ada!"` and does not `print` to the terminal.
4. **The whole function:**

```
def greet(name):
  return f"Hello, {name}!"
```

Made by Thana

Personal worked-solution guide based on the APDR Computer Exercise 0 task sheet, rewritten in my own words. It is not an official FAU publication. Exercise 0 is practice; Exercises 1–2 are scored tasks. Use the samples and dataset from your own assigned repository where values are personalized. Made by **Thanate Amornjiraporn (Focus)**.